

GORAKHPUR, India

WMO: 423790

Lat: 26.740N

Lon: 83.450E

Elev: 79

StdP: 100.38

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.6	8.9	1.9	4.4	24.4	3.8	5.0	19.7	4.1	19.2	3.3	17.5	0.2	270	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.6	40.8	22.9	39.4	22.9	37.9	23.1	29.1	33.0	28.3	32.8	27.9	32.4	2.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.8	25.5	30.5	27.3	23.3	30.5	26.7	22.5	30.6	96.7	32.6	92.7	33.1	90.4	32.4	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.1	3.9	3.1	5.2	41.9	1.6	1.4	4.1	42.9	3.1	43.8	2.2	44.5	1.1	45.6		
			4.1	29.4	1.8	0.9	2.9	30.1	1.8	30.6	0.9	31.1	-0.4	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.5	14.8	19.4	24.6	29.8	31.4	31.6	29.8	29.6	29.1	26.8	22.1	16.9	(6)
(7)		DBStd	6.05	2.79	2.41	2.71	2.27	2.26	2.49	1.65	1.28	1.54	2.03	1.91	2.69	(7)
(8)		HDD10.0	3	3	0	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	183	110	14	0	0	0	0	0	0	0	0	1	58	(9)
(10)		CDD10.0	5666	153	263	452	596	663	649	613	609	573	520	363	213	(10)
(11)		CDD18.3	2804	2	43	194	346	405	399	354	351	323	262	113	13	(11)
(12)		CDH23.3	35147	11	298	1976	4776	5975	5872	4605	4486	3921	2470	682	76	(12)
(13)		CDH26.7	18116	0	45	931	2893	3676	3528	2167	2043	1690	979	162	2	(13)
(14)	Wind	WSAvg	0.9	0.6	0.8	1.1	1.3	1.4	1.3	1.1	1.1	0.9	0.5	0.3	0.4	(14)
(15)	Precipitation	PrecAvg	1089	18	17	9	5	48	140	289	277	198	50	3	4	(15)
(16)		PrecMax	1951	72	64	60	18	151	496	688	618	524	237	52	54	(16)
(17)		PrecMin	626	1	1	2	0	3	0	41	83	1	0	0	0	(17)
(18)		PrecStd	321	19	18	13	5	37	129	165	135	113	71	10	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	30.1	36.7	41.1	42.8	42.5	36.3	35.2	35.0	34.1	31.0	26.8	(19)
(20)		0.4%	MCWB	16.3	19.6	20.6	20.9	22.2	24.0	27.4	28.3	26.8	25.1	20.7	17.6	(20)
(21)		2%	DB	23.3	28.1	34.7	39.7	40.9	40.9	34.9	34.1	34.1	33.1	29.9	25.3	(21)
(22)		2%	MCWB	15.6	18.7	20.2	20.3	22.5	24.7	27.5	28.0	26.8	25.0	20.4	16.9	(22)
(23)		5%	DB	22.0	26.6	33.3	38.4	39.4	39.0	33.9	33.4	33.4	32.2	28.7	24.0	(23)
(24)		5%	MCWB	15.2	18.0	19.7	20.5	23.0	25.4	27.4	27.7	26.7	24.6	20.1	16.7	(24)
(25)		10%	DB	20.5	25.2	31.8	37.0	37.9	37.0	33.0	32.7	32.5	31.3	27.3	22.4	(25)
(26)		10%	MCWB	14.7	17.4	19.5	20.8	23.6	26.3	27.3	27.5	26.5	24.2	19.7	16.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	21.7	24.4	25.7	28.0	29.2	30.0	30.9	29.9	27.6	23.2	19.7	(27)
(28)		0.4%	MCD B	22.9	27.4	32.6	35.2	36.8	35.7	32.0	33.3	31.3	31.5	27.6	24.3	(28)
(29)		2%	WB	16.9	20.4	22.5	24.8	27.0	28.5	28.7	29.4	28.0	26.8	22.0	18.2	(29)
(30)		2%	MCD B	21.5	26.4	30.8	34.6	35.5	35.1	32.0	31.7	31.8	31.1	27.3	23.3	(30)
(31)		5%	WB	16.0	19.1	21.3	24.0	26.4	28.0	28.1	28.2	27.4	26.1	21.3	17.3	(31)
(32)		5%	MCD B	20.4	24.9	29.9	33.6	34.6	34.0	32.1	31.8	31.5	30.2	26.5	22.1	(32)
(33)		10%	WB	15.1	18.1	20.4	23.2	25.7	27.5	27.6	27.7	27.0	25.5	20.5	16.5	(33)
(34)		10%	MCD B	19.5	23.5	29.2	32.7	33.8	33.3	31.7	31.5	31.3	29.4	25.7	21.2	(34)
(35)	Mean Daily Temperature Range	MDBR	9.6	11.6	13.4	14.1	11.7	8.6	5.2	5.4	6.1	9.3	12.2	10.6	(35)	
(36)		5% DB	MCD BR	12.8	12.8	15.1	15.8	14.5	12.4	6.6	6.6	7.3	9.8	12.8	13.0	(36)
(37)		5% DB	MCWBR	6.7	5.7	4.9	5.0	4.6	3.8	2.3	2.8	2.5	3.6	5.3	6.3	(37)
(38)		5% WB	MCD BR	11.2	11.7	13.6	13.0	10.9	8.9	6.1	6.1	6.5	7.8	11.0	11.4	(38)
(39)		5% WB	MCWBR	6.4	6.2	5.7	5.3	4.4	3.3	2.9	3.2	3.0	3.5	5.1	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.561	0.535	0.528	0.611	0.699	0.719	0.633	0.583	0.574	0.625	0.657	0.620	(40)	
(41)		taud	1.725	1.832	1.878	1.740	1.612	1.609	1.803	1.949	1.938	1.749	1.615	1.637	(41)	
(42)		Ebn at Noon	663	735	774	723	661	644	701	735	727	651	578	586	(42)	
(43)		Edh at Noon	204	197	198	233	265	264	218	187	184	211	227	216	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.92	4.36	5.73	6.49	6.37	5.43	4.47	4.59	4.57	4.53	3.68	2.89	(44)	
(45)		RadStd	0.42	0.31	0.19	0.30	0.36	0.50	0.40	0.26	0.33	0.29	0.20	0.36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page